

MACHINE LEARNING II

DYNAMIC PROGRAMMING, ITERATED CONDITIONAL MODE

Exercise 1.

a) A square is divided into $n \times n$ fields. There are costs $c(x,y)$ for each field (x,y) . Consider all paths that

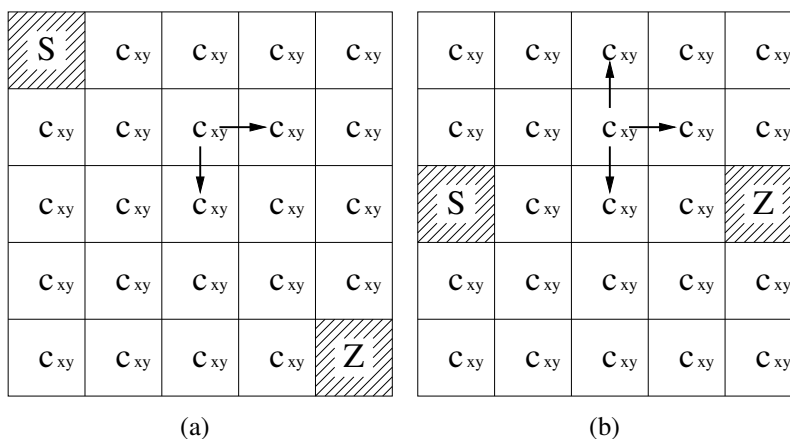
- (1) connect the leftmost top field (marked as “S”) with the rightmost bottom one (marked as “Z”) and
- (2) starting from a field continue to the right or to the lower neighbour (see Fig. (a)).

The total cost for a path is the sum of the visited field costs. Design an efficient algorithm to find the path of the minimal total cost.

b) Change the problem as follows:

- (1) The path connects the leftmost middle field with the rightmost middle field.
- (2) Starting from an actual field it can continue to the right, to the upper or to the lower neighbour (see Fig. (b)).

Design an efficient algorithm also for this modified case.



Exercise 2. The task is to “cover” an interval of the length n by a couple of sub-intervals. Each sub-interval is characterized by its start position (let us denote it by b – “begin”) and its end position (denoted by e – “end”). There are costs for all possible sub-intervals (b,e) (i.e. $n(n-1)/2$ numbers in total). The “covering” is a set of subintervals that (i) there is no gap between sub-intervals and (ii) sub-intervals may not overlap. The cost of

a covering is the summed costs of all used sub-intervals. Design an efficient algorithm to find a covering of the minimal total cost.

Exercise 3. Consider a “usual” task of Dynamic Programming to find the label sequence of the minimal total energy. Remember that in general the time complexity of the Dynamic Programming is $O(nK^2)$, where n is the length and K is the number of labels. Assume that there are

- a) K processors
- b) $n/2$ processors
- c) $n/2 \cdot K^2$ processors

at our disposition to perform the computations. Which computations can be parallelized? Which “wall clock” time complexity can be reached?

Hint: Especially for b) and c) remember the variant of Dynamic Programming for cycles discussed at the lecture.

Exercise 4. Consider one “elementary step” of the ICM-Algorithm, i.e. the choice of the best label in a node given the fixed rest:

$$x_i = \arg \min_{k \in K} \left[\theta_i(k) + \sum_{j: i,j \in \mathcal{E}} \theta_{ij}(k, x_j) \right].$$

The time complexity of this step is $O(m|K|)$ in general, where m is the number of nodes that are linked with i by an edge and $|K|$ is the number of labels.

How to improve the time complexity, if the functions θ_{ij} are:

- a) the Potts model $\theta_{ij}(k, k') = a_{ij} \cdot \delta(k \neq k')$
- b) quadratic distance between labels $\theta_{ij}(k, k') = a_{ij} \cdot (k - k')^2$
- c) hard restrictions $\theta_{ij}(k, k') = \begin{cases} 0 & \text{if } |k - k'| < \Delta \\ \infty & \text{otherwise} \end{cases}$
- d) a modified version of c) $\theta_{ij}(k, k') = \begin{cases} a & \text{if } |k - k'| < \Delta \\ b & \text{otherwise} \end{cases}$

Note that in b) – d) the labels are ordered. Hence, expressions like e.g. $(k - k')^2$ are well-defined.